Music Playlist Manager

Doubly Linked List - Source Code and Output

Data Structures and Algorithms [R25-EC-PCC-202] | CCE Case Study, Semester III, Oct. 2026

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1. Source code (main.cpp)

```
#include <iostream>
#include <string>
using namespace std;

class Node
{
public:
    string title, artist;
    Node *prev, *next;

    Node(string t, string a)
    {
        title = t;
        artist = a;
        prev = next = NULL;
    }
};

class Playlist
{
    Node *head, *current;

public:
    Playlist()
    {
        head = current = NULL;
    }

    void addSong()
    {
        string title, artist;

        cin.ignore();
        cout << "Enter song title: ";
        getline(cin, title);
        cout << "Enter artist name: ";
        getline(cin, artist);

        Node *p = new Node(title, artist);

        if (head == NULL)
        {
            head = current = p;
        }
        else
        {
            Node *q = head;

            while (q->next != NULL)
                q = q->next;

            q->next = p;
            p->prev = q;
        }

        cout << "Song added successfully!\n";
    }

    void deleteSong()
    {
        if (head == NULL)
        {
            cout << "Playlist is empty.\n";
            return;
        }

        string title;
```

```
        cin.ignore();
        cout << "Enter song title to delete: ";
        getline(cin, title);

        Node *p = head;

        while (p != NULL && p->title != title)
            p = p->next;

        if (p == NULL)
        {
            cout << "Song not found.\n";
            return;
        }

        if (p->prev != NULL)
            p->prev->next = p->next;
        else
            head = p->next;

        if (p->next != NULL)
            p->next->prev = p->prev;

        if (current == p)
            current = head;

        delete p;
        cout << "Song deleted successfully!\n";
    }

    void display()
    {
        Node *q = head;

        if (q == NULL)
        {
            cout << "Playlist is empty.\n";
            return;
        }

        while (q != NULL)
        {
            cout << q->title << " - " << q->artist;

            if (q == current)
                cout << " [CURRENT]";

            cout << endl;
            q = q->next;
        }
    }

    void reverseDisplay()
    {
        Node *q = head;

        if (q == NULL)
        {
            cout << "Playlist is empty.\n";
            return;
        }

        while (q->next != NULL)
            q = q->next;

        while (q != NULL)
        {
            cout << q->title << " - " << q->artist << endl;
            q = q->prev;
        }
    }

    void playCurrent()
    {
        if (current == NULL)
            cout << "Playlist is empty.\n";
        else
            cout << "Playing: " << current->title
                 << " - " << current->artist << endl;
```

```
147     }
148
149     void nextSong()
150     {
151         if (current == NULL)
152             cout << "Playlist is empty.\n";
153         else if (current->next == NULL)
154             cout << "Already at the last song.\n";
155         else
156         {
157             current = current->next;
158             playCurrent();
159         }
160     }
161
162     void previousSong()
163     {
164         if (current == NULL)
165             cout << "Playlist is empty.\n";
166         else if (current->prev == NULL)
167             cout << "Already at the first song.\n";
168         else
169         {
170             current = current->prev;
171             playCurrent();
172         }
173     }
174
175     void playAll()
176     {
177         Node *q = head;
178
179         if (q == NULL)
180         {
181             cout << "Playlist is empty.\n";
182             return;
183         }
184
185         while (q != NULL)
186         {
187             cout << "Playing: " << q->title
188                  << " - " << q->artist << endl;
189             q = q->next;
190         }
191     }
192
193     void repeatAll()
194     {
195         int n;
196
197         if (head == NULL)
198         {
199             cout << "Playlist is empty.\n";
200             return;
201         }
202
203         cout << "Enter number of repetitions: ";
204         cin >> n;
205
206         for (int i = 1; i <= n; i++)
207         {
208             cout << "Repetition " << i << endl;
209             playAll();
210         }
211     }
212
213     void showCurrent()
214     {
215         playCurrent();
216     }
217 };
218
219 int main()
220 {
221     Playlist p;
222     int choice;
223
224     do
225     {
```

```
        cout << "\n--- MUSIC PLAYLIST MANAGER ---\n";
        cout << "1. Add Song\n";
        cout << "2. Delete Song\n";
        cout << "3. Display Playlist\n";
        cout << "4. Display in Reverse\n";
        cout << "5. Play Current Song\n";
        cout << "6. Play Next Song\n";
        cout << "7. Play Previous Song\n";
        cout << "8. Play Complete Playlist\n";
        cout << "9. Repeat All\n";
        cout << "10. Show Current Song\n";
        cout << "11. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        switch (choice)
        {
        case 1:
            p.addSong();
            break;
        case 2:
            p.deleteSong();
            break;
        case 3:
            p.display();
            break;
        case 4:
            p.reverseDisplay();
            break;
        case 5:
            p.playCurrent();
            break;
        case 6:
            p.nextSong();
            break;
        case 7:
            p.previousSong();
            break;
        case 8:
            p.playAll();
            break;
        case 9:
            p.repeatAll();
            break;
        case 10:
            p.showCurrent();
            break;
        case 11:
            cout << "Thank you!\n";
            break;
        default:
            cout << "Invalid choice.\n";
        }

    } while (choice != 11);

    return 0;
}
```

2. Program output

```
PS C:\Users\prati\Desktop\DSA project> .\main

--- MUSIC PLAYLIST MANAGER ---
1. Add Song
2. Delete Song
3. Display Playlist
4. Display in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter choice: 1
Enter song title: Kesariya
Enter artist name: Arijit singh
Song added successfully!

--- MUSIC PLAYLIST MANAGER ---
1. Add Song
2. Delete Song
3. Display Playlist
4. Display in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter choice: 1
Enter song title: Tum Hi Ho
Enter artist name: Arijit singh
Song added successfully!

--- MUSIC PLAYLIST MANAGER ---
1. Add Song
2. Delete Song
3. Display Playlist
4. Display in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter choice: 1
Enter song title: Channa Mereya
Enter artist name: Arijit Singh
Song added successfully!

--- MUSIC PLAYLIST MANAGER ---
1. Add Song
2. Delete Song
3. Display Playlist
4. Display in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter choice: 3
Kesariya - Arijit singh [CURRENT]
Tum Hi Ho - Arijit singh
Channa Mereya - Arijit Singh

--- MUSIC PLAYLIST MANAGER ---
1. Add Song
2. Delete Song
3. Display Playlist
4. Display in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
```

```
10. Show Current Song
11. Exit
Enter choice: 4
Channa Mereya - Arijit Singh
Tum Hi Ho - Arijit singh
Kesariya - Arijit singh

--- MUSIC PLAYLIST MANAGER ---
1. Add Song
2. Delete Song
3. Display Playlist
4. Display in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter choice: 5
Playing: Kesariya - Arijit singh

--- MUSIC PLAYLIST MANAGER ---
1. Add Song
2. Delete Song
3. Display Playlist
4. Display in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter choice: 6
Playing: Tum Hi Ho - Arijit singh

--- MUSIC PLAYLIST MANAGER ---
1. Add Song
2. Delete Song
3. Display Playlist
4. Display in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter choice: 7
Playing: Kesariya - Arijit singh

--- MUSIC PLAYLIST MANAGER ---
1. Add Song
2. Delete Song
3. Display Playlist
4. Display in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter choice: 8
Playing: Kesariya - Arijit singh
Playing: Tum Hi Ho - Arijit singh
Playing: Channa Mereya - Arijit Singh

--- MUSIC PLAYLIST MANAGER ---
1. Add Song
2. Delete Song
3. Display Playlist
4. Display in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter choice: 9
```

```
Enter number of repetitions: 5
Repetition 1
Playing: Kesariya - Arijit singh
Playing: Tum Hi Ho - Arijit singh
Playing: Channa Mereya - Arijit Singh
Repetition 2
Playing: Kesariya - Arijit singh
Playing: Tum Hi Ho - Arijit singh
Playing: Channa Mereya - Arijit Singh
Repetition 3
Playing: Kesariya - Arijit singh
Playing: Tum Hi Ho - Arijit singh
Playing: Channa Mereya - Arijit Singh
Repetition 4
Playing: Kesariya - Arijit singh
Playing: Tum Hi Ho - Arijit singh
Playing: Channa Mereya - Arijit Singh
Repetition 5
Playing: Kesariya - Arijit singh
Playing: Tum Hi Ho - Arijit singh
Playing: Channa Mereya - Arijit Singh

--- MUSIC PLAYLIST MANAGER ---
1. Add Song
2. Delete Song
3. Display Playlist
4. Display in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter choice: 10
Playing: Kesariya - Arijit singh

--- MUSIC PLAYLIST MANAGER ---
1. Add Song
2. Delete Song
3. Display Playlist
4. Display in Reverse
5. Play Current Song
6. Play Next Song
7. Play Previous Song
8. Play Complete Playlist
9. Repeat All
10. Show Current Song
11. Exit
Enter choice: 11
Thank you!
PS C:\Users\prati\Desktop\DSA project>
```